

NAME (PRINT):

Last/Surname

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STUDENT #:

SIGNATURE:

**UNIVERSITY OF TORONTO MISSISSAUGA
DECEMBER 2019 FINAL EXAMINATION
CSC343H5F
Introduction to Databases
Michael Liut
Duration - 3 hours
Aids: None**

The University of Toronto Mississauga and you, as a student, share a commitment to academic integrity. You are reminded that you may be charged with an academic offence for possessing any unauthorized aids during the writing of an exam. Clear, sealable, plastic bags have been provided for all electronic devices with storage, including but not limited to: cell phones, smart watches, SMART devices, tablets, laptops, and calculators. Please turn off all devices, seal them in the bag provided, and place the bag under your desk for the duration of the examination. You will not be able to touch the bag or its contents until the exam is over.

If, during an exam, any of these items are found on your person or in the area of your desk other than in the clear, sealable, plastic bag, you may be charged with an academic offence. A typical penalty for an academic offence may cause you to fail the course.

*Please note, once this exam has begun, you **CANNOT** re-write it.*

To pass this course you are required to obtain at least 40% on this examination.

Additional instructions are located on page 3 of this exam booklet. You must read them prior to starting the exam. Failure to do so could result in mark reductions.

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Name: _____

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CSC 343 H5F

Duration: 3 hours
December 2019

Final Examination

Instructor: Dr. Michael Liut
University of Toronto Mississauga

Read all instructions before starting the exam

This test paper includes 23 pages and a total of 23 questions; each question requires a written answer, even justification for the true-or-false. You are responsible for ensuring that your copy of the paper is complete. Bring any discrepancy to the attention of your invigilator.

Special Instructions:

1. The answers to all questions must be written in the space provided (i.e. immediately following the question). Written answers are to be clear and concise; illegible solutions will not be marked and awarded a grade of 0.
2. Read all questions before starting the exam - some are easier than others.
3. **This is a closed book exam.** No memory aids or textbooks of any kind are allowed during the test. Supporting pages have been appended to this exam.
4. The students are not allowed to be involved in any communication of any kind concerning the questions and answers of this exam with anybody except the invigilator(s) or the instructor(s). Any attempt of such communication will be considered a case of academic dishonesty.
5. No unauthorized scrap paper is allowed to be used. The invigilator(s) will supply every student with needed scrap paper when asked.
6. Documents to be returned: this questionnaire (i.e. the exam) and, if used, all scrap paper. All documents must bear the student's name and student's identification number (i.e. UtorID). Only the face page of the questionnaire needs to bear the student's name and number.
7. On this page, to the right of this line, draw an animal of your choice and give it a name. By doing so you will be awarded with one additional point on this exam.

question	mark	out of
1–18		27
19		28
20		12
21		4
22		3
23		10
total		84

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Questions 1 – 18 are True-or-False questions. You are required to illustrate the validity by an example, or disprove the validity by a counter-example

For each question only select one answer (i.e. True or False). To select an answer, you must circle the word “True” or the word “False”. Incorrect or missing answers earn a mark of 0. Therefore, do not leave any questions blank. For full marks you must justify (prove/disprove) your answer.

Question 1 [1.5 marks]

Aside from buying a faster system, throughput can be increased in three ways:

1. By locking the smallest sized objects possible.
2. By reducing the time that transactions hold locks.
3. By reducing *hot spots*¹.

- A. True.
- B. False.

Question 2 [1.5 marks]

Given the four levels of isolation², read committed is susceptible³ of dirty reads, as a read committed transaction never makes any lock request.

- A. True.
- B. False.

Question 3 [1.5 marks]

Set union is idempotent (i.e. $S \cup S = S$), while under bag law it is not (i.e. $S \cup S \neq S$). Thus, in general, we may assume that intersection is also NOT idempotent (i.e. $S \cap S \neq S$).

- A. True.
- B. False.

¹real definition: a hot spot is a database object that is frequently accessed and modified.

²read uncommitted, read committed, repeatable read, and serializable.

³real definition: capable of.

Question 4 [1.5 marks]

Given two relations, **Emps** and **WorksIn**, if the following SQL Query is executed, then its result is undoubtedly the **Output** relation.

Emps:

Emp	Sal	Mgr
Xin	100	Pat
Jen	200	Jen
Pat	100	Meg
Meg	150	Meg
Joe	110	Ami
Ami	200	Bob

WorksIn:

Dept	Emp
Toy	Xin
Car	Bob
Car	Meg
Art	Pat
Art	Meg

SQL Query

```
SELECT *
FROM Emps,
    (select Emps.emp, max(sal) as sal
     from Emps, WorksIn
     where Emps.emp = WorksIn.emp
     group by Emps.emp) as foo
WHERE Emps.mgr = foo.emp
```

Output:

Emp	Sal	Mgr	Emp	Sal
Xin	100	Pat	Pat	100
Pat	100	Meg	Meg	150
Meg	150	Meg	Meg	150

- A. True.
- B. False.

Question 5 [1.5 marks]

Given the two relations in Question 4, **Emps** and **WorksIn**, if the following SQL Query is executed, then its result is undoubtedly the **Output** relation.

SQL Query

```
SELECT E1.Emp
FROM Emps E1 full outer join Emps E2 on E1.Emp = E2.Mgr
GROUP BY E1.Emp, E2.Emp
HAVING sum(E1.Sal) >= sum(E2.Sal)
ORDER BY E1.Emp
```

Output:

Emp
Ami
Jen
Meg
Meg
Pat

- A. True.
- B. False.

Question 6 [1.5 marks]

Let's assume that our DBMS supports both static and dynamic tree indexing. A static index (e.g. B+ tree) without overflow pages is said to be faster than a dynamic index (e.g. ISAM⁴) on updates only.

- A. True.
- B. False.

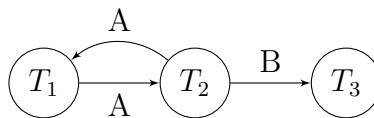
Question 7 [1.5 marks]

In scheduling, specifically database transactions, Avoiding Cascading Aborts and Serializability are noninherently-blocking properties, while Recoverability, Strictness and Two-Phase Locking are inherently-blocking properties.

- A. True.
- B. False.

Question 8 [1.5 marks]

The following sequence of transactions: $R_1(A)$, $R_2(A)$, $R_2(B)$, $R_3(B)$, $R_3(A)$, $W_1(A)$, $W_2(A)$, $W_3(B)$ is represented below by its precedence graph. This graph tells us that this schedule is conflict serializable.



- A. True.
- B. False.

Question 9 [1.5 marks]

The two ACID properties that a recovery manager of a DBMS is responsible for ensuring are consistency and durability. After a crash the DBMS must be brought back into a consistent state and ensure that all committed transactions survive and persist.

- A. True.
- B. False.

⁴Indexed Sequential Access Method: in contrast to a B+ tree index, empty spaces in index pages are never filled.

Question 10 [1.5 marks]

Given the following relation R :

A	B	C
1	2	3
4	2	3
5	3	3

the functional dependencies $A \rightarrow B$, $B \rightarrow C$, and $BC \rightarrow A$ all hold.

- A. True.
- B. False.

Question 11 [1.5 marks]

Aggregation is a term used to describe one of the fundamental uses of DBMSs. It also allows us to treat a relationship set as an entity-set for the purposes of participation in (other) relationships.

- A. True.
- B. False.

Question 12 [1.5 marks]

A candidate key is a minimal key that allows us to uniquely identify all tuples in a relation. The database administrator/architect selects one of those keys, as there can be many, and makes a decision on which will be the great key by arbitrary or strategic selection.

- A. True.
- B. False.

Question 13 [1.5 marks]

Virtual views are constructed and stored. Further, virtual views may be periodically updated depending on their query definition.

- A. True.
- B. False.

Question 14 [1.5 marks]

There exists a scenario, when searching for a data record, in which a secondary index is more practical to use than a primary index.

- A. True.
- B. False.

Question 15 [1.5 marks]

Fine grained locking causes a lower degree of concurrency and increases locking overhead, while coarse grained locking causes a higher degree of concurrency.

- A. True.
- B. False.

Question 16 [1.5 marks]

A weak entity-set E has a relationship with a supporting entity-set R. The relationship between E and R must be one-to-one.

- A. True.
- B. False.

General Commentary

SQL supports four levels of isolation and two access-modes(read-only and read-write), for a total of eight possible combinations. Each combination implicitly defines a class of transactions.

Question 17 [1.5 marks]

Given the general commentary above, if a schedule was generated by the execution of several SQL transactions, each of which has a serializable isolation level it is guaranteed that the schedule would be conflict-serializable, serializable, and recoverable.

- A. True.
- B. False.

Question 18 [1.5 marks]

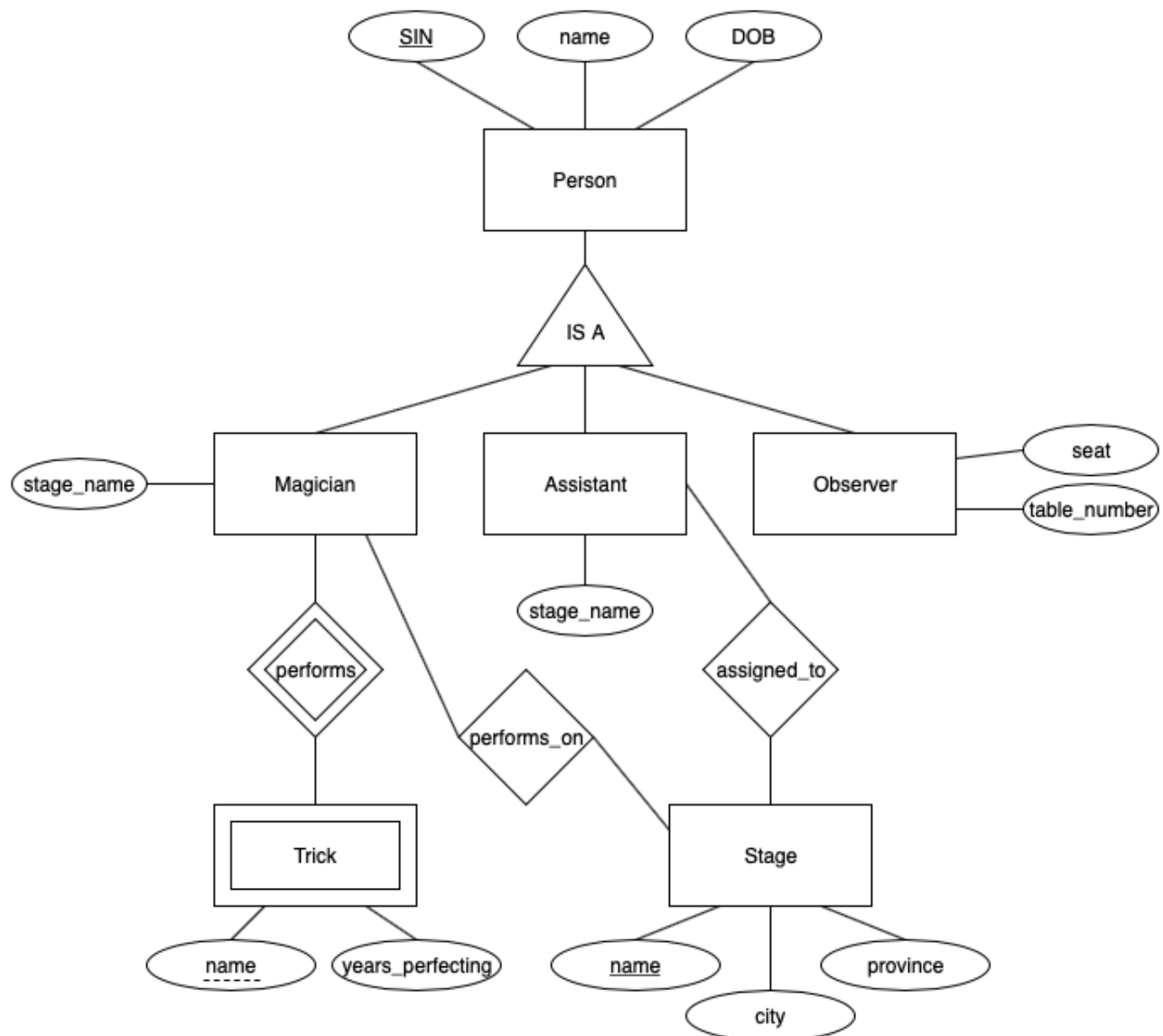
Given the general commentary above, if we are in the isolation state read committed and access-mode read-write, obtains locks on individual objects following the S2PL protocol.

- A. True.
- B. False.

Questions 19 – 23 are questions that require a written answer. The answers are to be written in the space provided below each question.

Question 19 [28 marks]

Without knowing the specifics of the multiplicity, you are given a rough draft of an Entity-Relationship Diagram (ERD) from a member of your design team:



Task I [8 marks]

Given the ERD on the previous page, you are to directly⁵ correct⁶ the ERD to reflect the following:

1. Every magician must perform exactly 2 tricks or no tricks at all.
2. Each observer can only visit a stage once. Observers may visit many different stages. Each visit will have an associated **entry_time:datetime**, signifying the date and time which the observer will be permitted to visit that stage. There can be several (or no) observers visiting any stage. Observers' attributes have no bearing on the visit.
3. Every assistant must be assigned to exactly one stage.

NOTE: you may not make any additional assumptions (i.e. you must take all avenues in to consideration) and if there is no valid way to demonstrate this in an ERD, then you must explain why and how you would validate this from a DDL perspective.

⁵you must write your answers directly on the ERD that was given, rather than drawing a new ERD.

⁶you must utilize Chen's Notation. A reference sheet has been provided towards the end of this exam.

Task II [20 marks]

For each of the below queries, write their corresponding SQL statements and their Relational Algebra counterpart.

- (a) Write a query that finds all magicians with one trick that they have been perfecting for more than three years.
- (b) Find all of David Copperfield's and Greg Frewin's (magicians) assistants who performed on a stage in Toronto, Ontario. You may assume: (1) this ERD only concerns itself with Canadian stages (hence only having a city and province); (2) an alternative key for stage is {city, name}; and (3) an assistant who is assigned to a stage has always been assigned there (dismiss any temporal component).

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Question 20 [12 marks]

Suppose you are given a relation R with a set of attributes $\alpha = \{A, B, C, D\}$. For each of the sets of functional dependencies (FDs) below, you are tasked with the following:

- i. identify the candidate keys for R ;
- ii. identify the best normal form⁷ that R satisfies; and
- iii. if R is not in BCNF, decompose it into a set of BCNF relations that preserve the functional dependencies.

Note: if you believe there are no candidate keys or there is no possible BCNF decomposition, you must state so and briefly explain why. You are required to show your work and justify your answers.

The set of functional dependencies⁸ $\mathcal{F}$ are:

1. $\{C \rightarrow D, C \rightarrow A, B \rightarrow C\}$
2. $\{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$

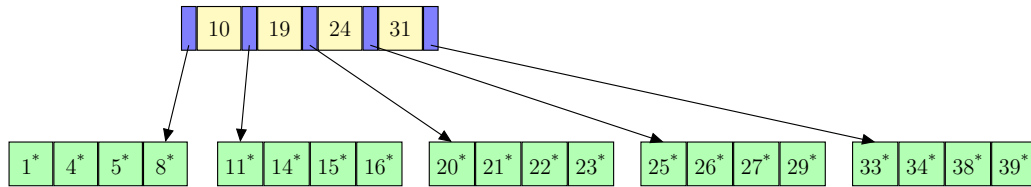
⁷Normal Forms: 1NF, 2NF, 3NF, and BCNF.

⁸each labeled set of FDs are independent of the next. They are the only FDs that hold for R .

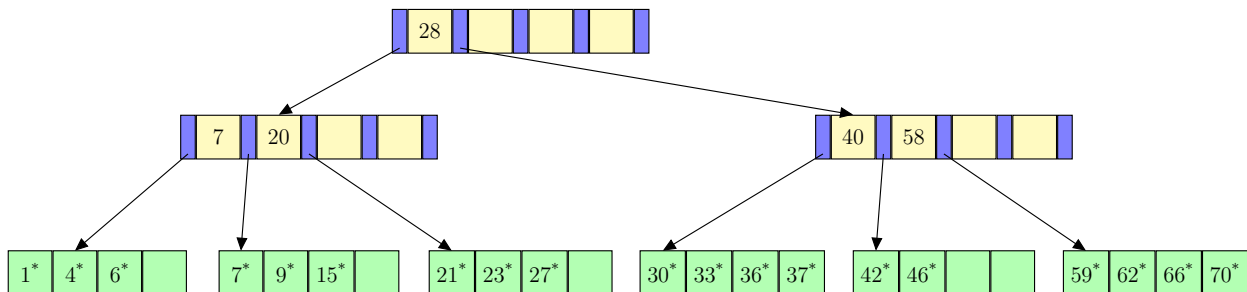
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Question 21 [4 marks] Given the following B+ Trees⁹, complete the following action:

- a. Insert 12, then 17, into the B+ Tree below. Provide the final result only.



- b. Delete 7 from the B+ Tree below. Provide the final result only.



⁹in the case of equality between a pointer and data record, the data record will fall directly to the right.

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Question 22 [3 marks] Consider the following data definition language:

```
CREATE TABLE Final(  
    A INT NOT NULL,  
    B INT NOT NULL,  
    C INT NOT NULL,  
    PRIMARY KEY (A),  
    FOREIGN KEY (B) REFERENCES Final(A) ON DELETE CASCADE ON UPDATE CASCADE,  
    FOREIGN KEY (C) REFERENCES Final(A) ON DELETE CASCADE ON UPDATE RESTRICT  
);
```

Now consider that the table `Final` has the following entries:

A	B	C
3	3	3
4	3	3
5	4	3

After the execution of the following SQL statement:

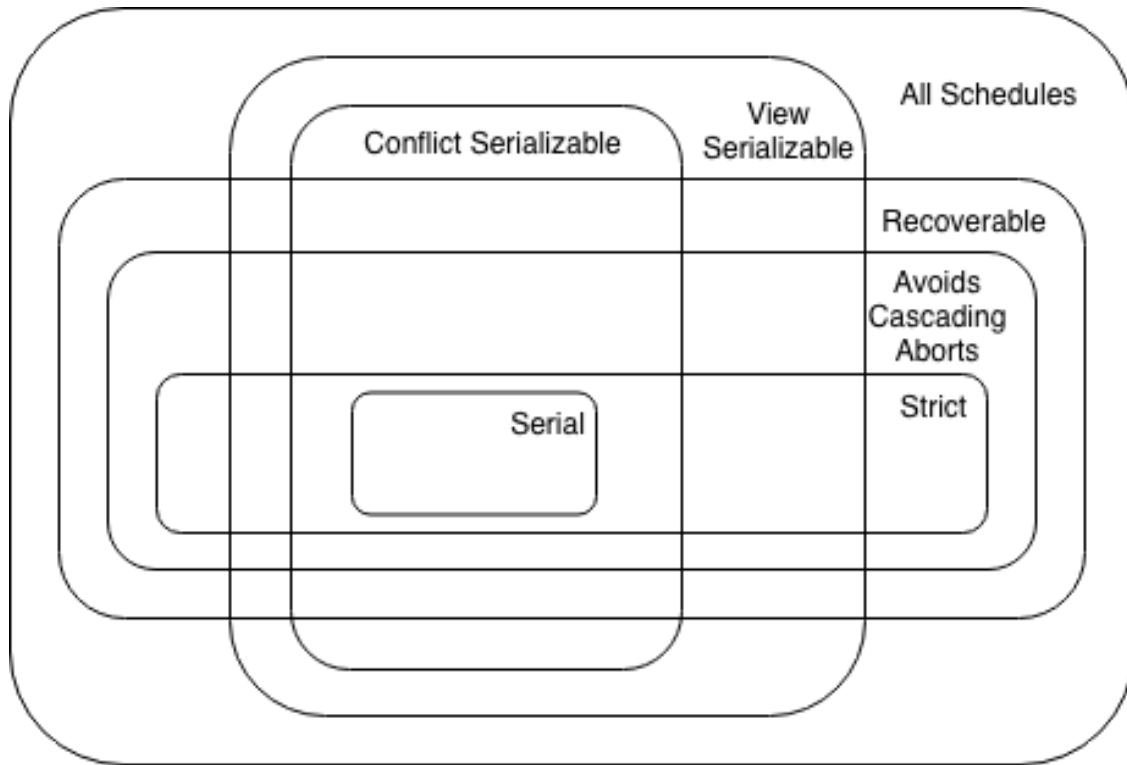
```
DELETE  
FROM Final  
WHERE A = 3;
```

What is the resulting table (draw it)? Explain your reasoning.

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Question 23 [10 marks]

Given the following Venn diagram:



and given the following (independent) sequence of transactions:

S_1 : $T_1:W(A)$, $T_2:R(A)$, $T_1:W(A)$, $T_2:Commit$, $T_1:Commit$

S_2 : $T_1:R(A)$, $T_2:W(A)$, $T_1:W(A)$, $T_3:R(A)$, $T_3:Commit$, $T_1:Commit$, $T_2:Commit$

S_3 : $T_1:W(A)$, $T_2:R(B)$, $T_1:R(B)$, $T_2:R(A)$, $T_1:Commit$, $T_2:Commit$

S_4 : $T_1:R(A)$, $T_2:W(A)$, $T_1:W(A)$, $T_2:Commit$, $T_1:Commit$

S_5 : $T_1:R(A)$, $T_2:W(A)$, $T_1:W(A)$, $T_2:Abort$, $T_1:Commit$

S_6 : $T_1:R(A)$, $T_2:R(A)$, $T_1:W(A)$, $T_1:Commit$, $T_2:W(A)$, $T_2:Commit$

using the sequence number (i.e. $S_{\#}$), mark on the Venn diagram where each of the sequences fall.

BONUS! You will be given one freebie, meaning that you can get one sequence wrong and still obtain full marks (i.e. you need only answer 5/6 questions correct)!

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REFERENCE SHEET

Chen's Notation

Participations Cardinality can be shown or hidden		Recursive Relationship Cardinality can be shown or hidden	
Entity	Entity	Mandatory	1 (0:1)
	Weak Entity	1	1 (1:1)
	Relationship	N	N (0:N)
	Identifying Relationship	1	N (1:N)
	Associative Entity	1	M (0:M)
Attribute	Attribute	Optional	1 (0:1)
	Key attribute	1	1 (1:1)
	Weak key attribute	N	N (0:N)
	Derived attribute	1	M (0:M)
	Multivalued attribute	1	M (1:M)

REFERENCE SHEET